Linear Discrimination

Classification $\Rightarrow$ $\mathcal{X} = \{(x_i, y_i)\}_{i=1}^{N}$

$x_i \in \mathbb{R}$ → univariate

$x_i \in \mathbb{R}^D$ → multivariate

$y_i \in \{1, 2, \dots, K\}$

$g_1(x)$
$g_2(x)$
$\vdots$
$g_K(x)$
} score functions

$$\hat{y}_{N+1} = \arg\max_{c=1}^{K} g_c(x_{N+1})$$

$$g_c(x) = p(x|y=c)\,Pr(y=c)$$

$Pr(y=c) = \frac{N_c}{N}$

N_c → # of data points in Class c

N → total # of data points

$p(x|y=c)$ → multivariate ($x_i \in \mathbb{R}^D$)

$p(x|y=c)$ → univariate ($x_i \in \mathbb{R}$)

$N(x_i; \mu_c, \sigma_c^2)$

$\hat{\mu}_c, \hat{\sigma}_c^2$ (Lecture 3)

$N(x_i; \mu_c, \Sigma_c)$

$\hat{\mu}_c, \hat{\Sigma}_c$ (Lecture 5)

$$g_c(x \mid W_c, w_c, w_{c0}) = x^T . \underbrace{W_c}_{D \times D} . x + \underbrace{w_c^T}_{D \times 1} . x + \underbrace{w_{c0}}_{1 \times 1}$$

total # of model parameters $= K\left[\frac{D(D+1)}{2} + D + 1\right]$

$a^2 + 2ab + b^2$

$$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} 1 & -5 \\ 7 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} a-5b \\ 7a+b \end{bmatrix}$$

$a^2 - 5ab + 7ab + b^2$

$a^2 + 2ab + b^2$

$$g_c(x \mid w_c, w_{c0}) = w_c^T . x + w_{c0}$$

total # of model parameters $= K[D+1]$

$$= \begin{bmatrix} w_{c1} & w_{c2} & \cdots & w_{cD} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_D \end{bmatrix} + w_{c0}$$

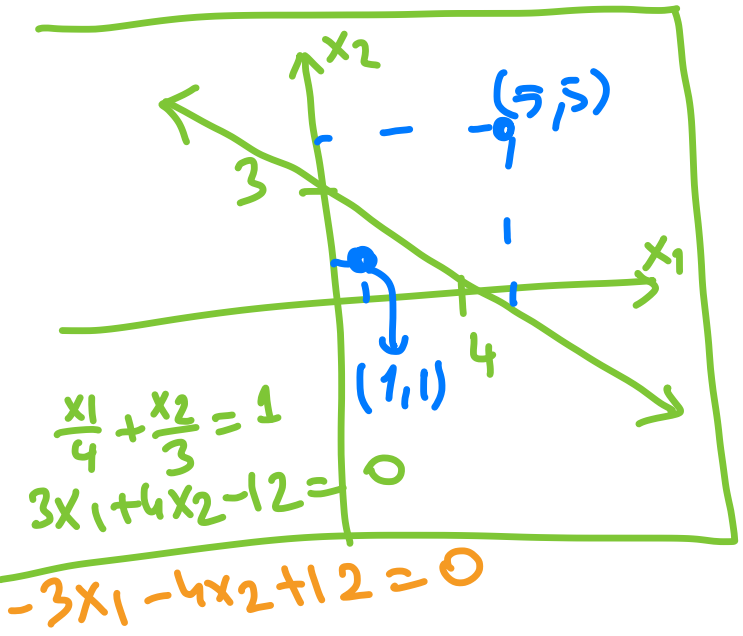
$$= \sum_{d=1}^{D} [w_{cd} . x_d] + w_{c0}$$

Binary Classification (K = 2)

$g_1(x)$, $g_2(x)$

if $g_1(x) > g_2(x) \Rightarrow \hat{y} = 1$

if $g_2(x) > g_1(x) \Rightarrow \hat{y} = 2$



if $g_1(x) - g_2(x) > 0 \Rightarrow \hat{y} = 1$

if $g_1(x) - g_2(x) < 0 \Rightarrow \hat{y} = 2$

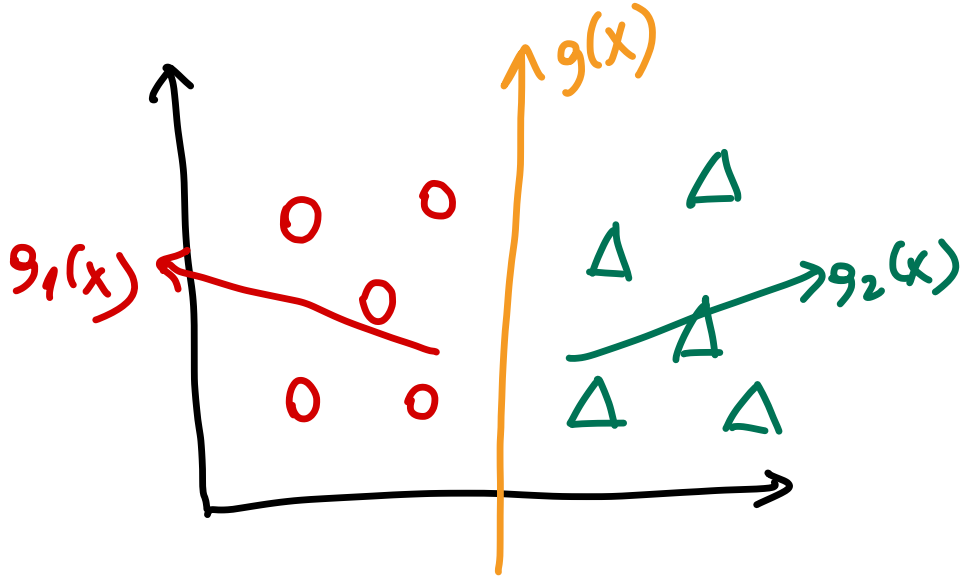
if $g(x) > 0 \Rightarrow \hat{y} = 1$

if $g(x) < 0 \Rightarrow \hat{y} = 2$

$g_1(x) = w_1^T . x + w_{10}$

$g_2(x) = w_2^T . x + w_{20}$

$g_1(x) - g_2(x) = (w_1 - w_2)^T . x + (w_{10} - w_{20})$



Multiclass classification ($K > 2$)

$g_1(x)$
$g_2(x)$
$\vdots$
$g_K(x)$

$\hat{y} = \arg\max_{c=1}^{K} g_c(x)$

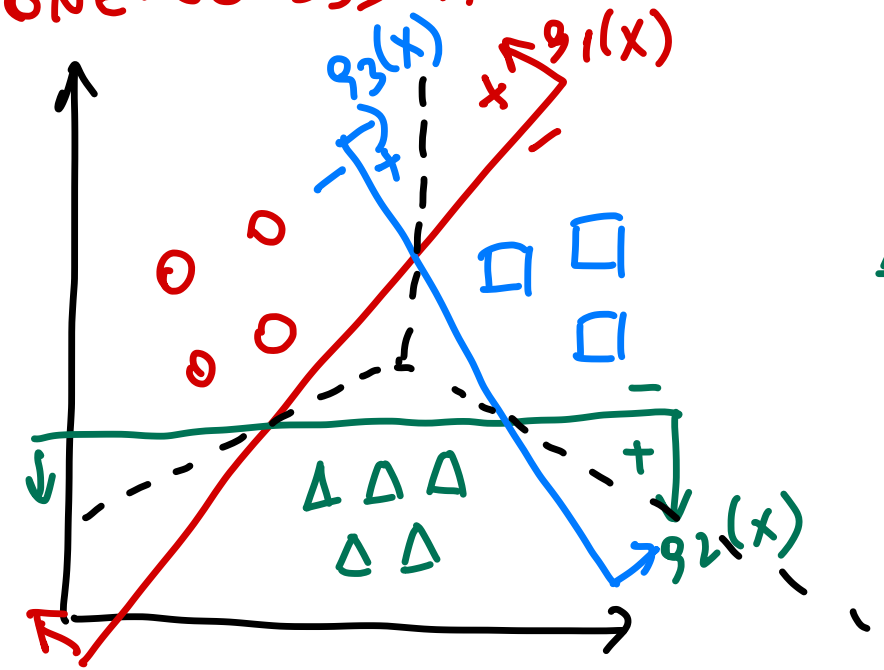
assume $K=3$

if $g_1(x) > g_2(x)$ & $g_1(x) > g_3(x)$ $\Rightarrow \hat{y} = 1$

if $g_2(x) > g_1(x)$ & $g_2(x) > g_3(x)$ $\Rightarrow \hat{y} = 2$

if $g_3(x) > g_1(x)$ & $g_3(x) > g_2(x)$ $\Rightarrow \hat{y} = 3$

ONE-VERSUS-ALL (OVA) APPROACH



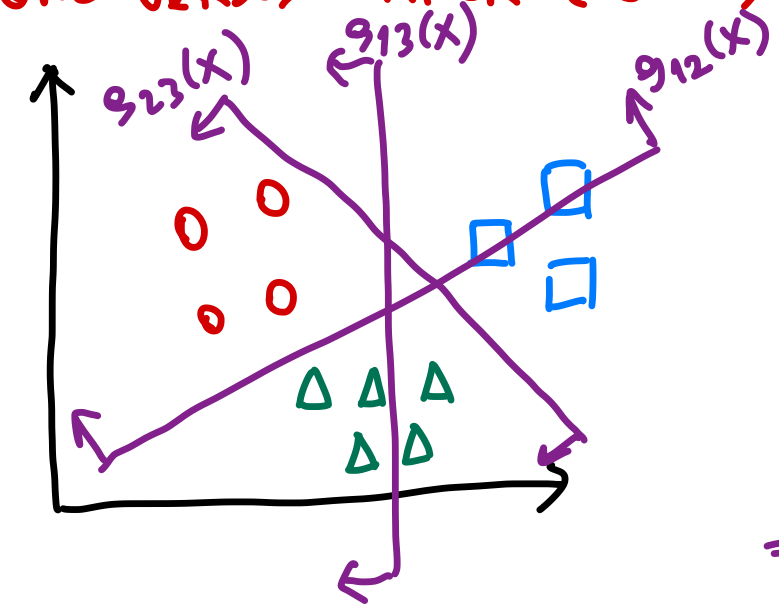
+	−
OOOO	ΔΔΔΔΔDDD $\Rightarrow N = 12$
ΔΔΔΔΔ	OOOO DDD $\Rightarrow N = 12$
DDD	OOOO ΔΔΔΔΔ $\Rightarrow N = 12$

of model parameters $= K(D+1)$

data set size $= N$

of score functions $= K$

ONE-VERSUS-OTHER (OVO) APPROACH



	+	−	
$g_{12}(x)$	OOOO	ΔΔΔΔΔ	$\Rightarrow N = 9$
$g_{13}(x)$	OOOO	□□□	$\Rightarrow N = 7$
$g_{23}(x)$	ΔΔΔΔΔ	□□□	$\Rightarrow N = 8$

$$\#\text{ of model parameters} = \frac{K(K-1)}{2}(D+1)$$

$$\text{data set size} = 2 \cdot \frac{N}{K}$$

$$\#\text{of score functions} = \frac{K(K-1)}{2}$$

X	1	2	3
$g_{12}(x)$	1	0	∅
$g_{13}(x)$	1	∅	0
$g_{23}(x)$	∅	0	1
# of wins $\Rightarrow$	2	0	1

→ pick the one with the highest # of wins.

1 : win

0 : loss

∅ : no participation

K=2

$$Pr(y=1|x) = \delta$$

$$Pr(y=2|x) = 1-\delta$$

$$\log\left(\frac{a\cdot b}{c\cdot d}\right) = \log\left(\frac{a}{c}\right) + \log\left(\frac{b}{d}\right)$$

if $\delta > 0.5 \Rightarrow \hat{y} = 1$

odds ratio ← if $\frac{\delta}{1-\delta} > 1 \Rightarrow \hat{y} = 1$

log-odds ratio ← if $\log\left[\frac{\delta}{1-\delta}\right] > 0 \Rightarrow \hat{y} = 1$

$$\log\left[\frac{Pr(y=1|x)}{Pr(y=2|x)}\right] = \log\left[\frac{p(x|y=1)\,Pr(y=1)\,/p(x)}{p(x|y=2)\,Pr(y=2)\,/p(x)}\right]$$

$$N(x;\mu_1,\Sigma_1) = \log\left[\frac{p(x|y=1)}{p(x|y=2)}\right] + \log\left[\frac{Pr(y=1)}{Pr(y=2)}\right]$$

$$N(x;\mu_2,\Sigma_2)$$

Assuming $\Sigma_1 = \Sigma_2 = \Sigma$ (shared covariance assumption)

$$p(x;\mu,\Sigma) = \frac{1}{\sqrt{(2\pi)^p|\Sigma|}} \cdot \exp\left[-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right]$$

$$= \log\left[\frac{\frac{1}{\sqrt{(2\pi)^p|\Sigma|}}\exp\left[-\frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1)\right]}{\frac{1}{\sqrt{(2\pi)^p|\Sigma|}}\exp\left[-\frac{1}{2}(x-\mu_2)^T\Sigma^{-1}(x-\mu_2)\right]}\right] + \log\left[\frac{Pr(y=1)}{Pr(y=2)}\right]$$

$$= -\frac{1}{2}(x-\mu_1)^T\Sigma^{-1}(x-\mu_1) + \frac{1}{2}(x-\mu_2)^T\Sigma^{-1}(x-\mu_2) + \log\left[\frac{Pr(y=1)}{Pr(y=2)}\right]$$

$$= -\frac{1}{2}x^T\Sigma^{-1}\cdot x + \mu_1^T\Sigma^{-1}\cdot x - \frac{1}{2}\mu_1^T\Sigma^{-1}\mu_1 + \frac{1}{2}x^T\Sigma^{-1}\cdot x - \mu_2^T\cdot\Sigma^{-1}\cdot x - \frac{1}{2}\mu_2^T\Sigma^{-1}\cdot\mu_2 + \dots$$

linear | constant | linear | constant + constant

$$(\mu_1-\mu_2)^T\cdot\Sigma^{-1}\cdot x = w^T$$

$$(A\cdot B)^T = B^T A^T$$

$$= W^T\cdot X + W_0$$

$$\hat{w} = \hat{\Sigma}^{-1}\cdot(\hat{\mu}_1-\hat{\mu}_2)$$

$$\hat{w}_0 = -\frac{1}{2}(\hat{\mu}_1+\hat{\mu}_2)^T\hat{\Sigma}^{-1}(\hat{\mu}_1-\hat{\mu}_2) + \log\left[\frac{\hat{Pr}(y=1)}{\hat{Pr}(y=2)}\right]$$